MODULE 03

C & C++ Programming

Build the programming fundamentals that make advanced technology easier

LEARN A SKILL • BUILD PROOF • CREATE YOUR NEXT OPPORTUNITY

Understand programming logic, memory, pointers and object-oriented programming instead of simply memorizing syntax.

WHO CAN JOIN?

Students, freshers, non-IT learners, coding-interview candidates and anyone who wants strong technical fundamentals.

WHY STUDENTS CHOOSE THIS PATH

- Beginner-friendly learning path — start with the fundamentals and progress step by step.
- Practical, project-focused training — learn by building instead of only watching lectures.
- Career-oriented skills — connect every topic to real applications and workplace use.
- Portfolio advantage — finish with projects you can discuss in interviews.
- A new direction is possible — your previous degree does not have to decide your entire career.

MODULE CONTENT

C fundamentals	Variables & data types
Operators	I/O
Conditions & loops	Functions
Arrays & strings	Pointers
Structures/unions/enums	Dynamic memory
Files	Preprocessor
Memory management	Stack vs heap
Function pointers	Bit manipulation
C++	Classes & objects
Constructors/destructors	Inheritance
Polymorphism	Encapsulation
Abstraction	Operator overloading
Templates	STL
Iterators	Smart pointers
Exceptions	Modern C++

HANDS-ON PROJECTS

- ATM Application
- Banking System
- Library Management
- Student Management
- Console Game

CAREER DIRECTIONS

C/C++ Developer • Junior Software Developer • Systems Programming Beginner • Competitive Programming Beginner

YOUR JOURNEY

Learn → Practice → Build → Create a portfolio → Prepare for interviews → Apply your skills in real-world situations.

Note: Training improves skills and employability; employment depends on individual preparation, portfolio quality, interviews and available opportunities.